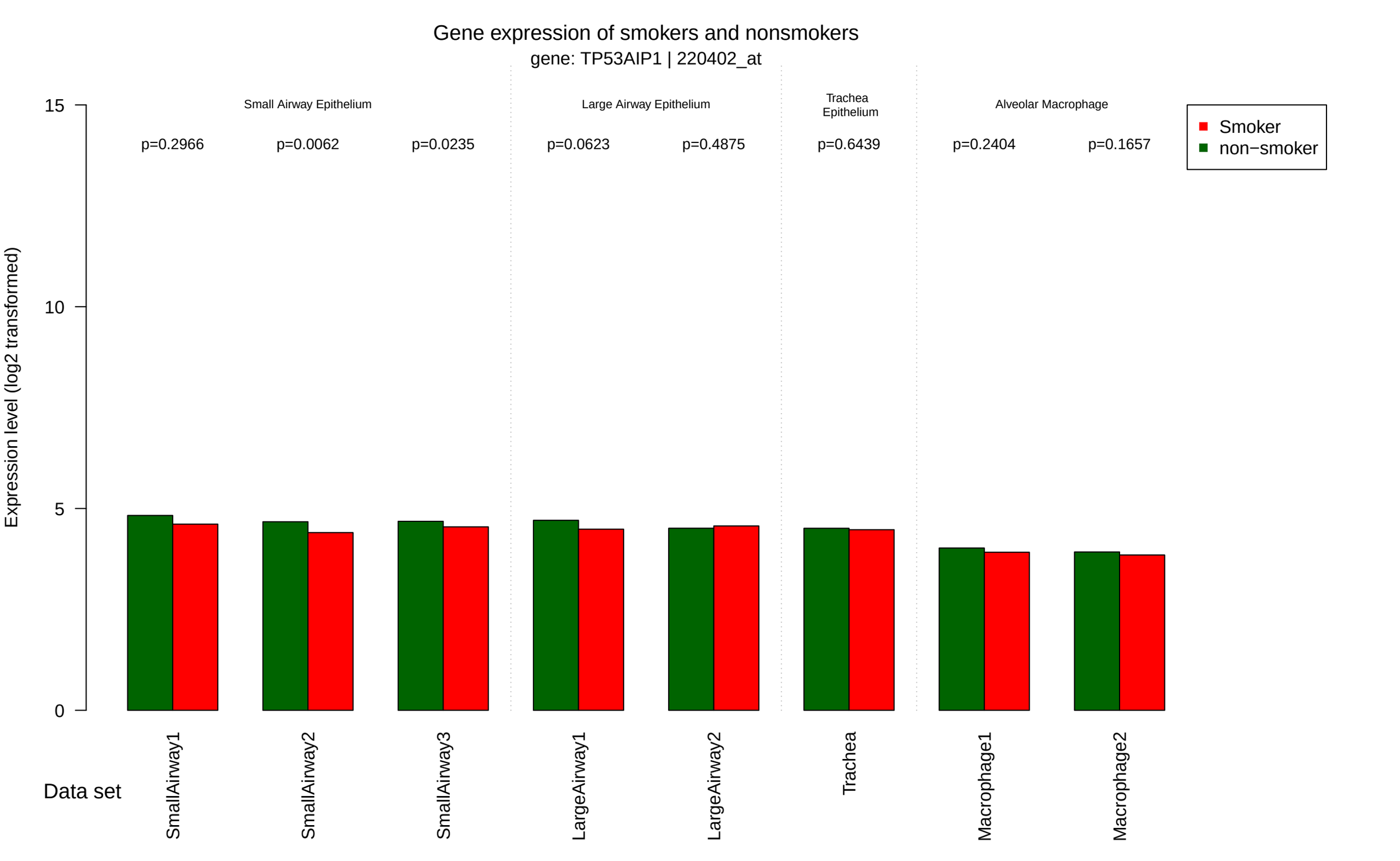
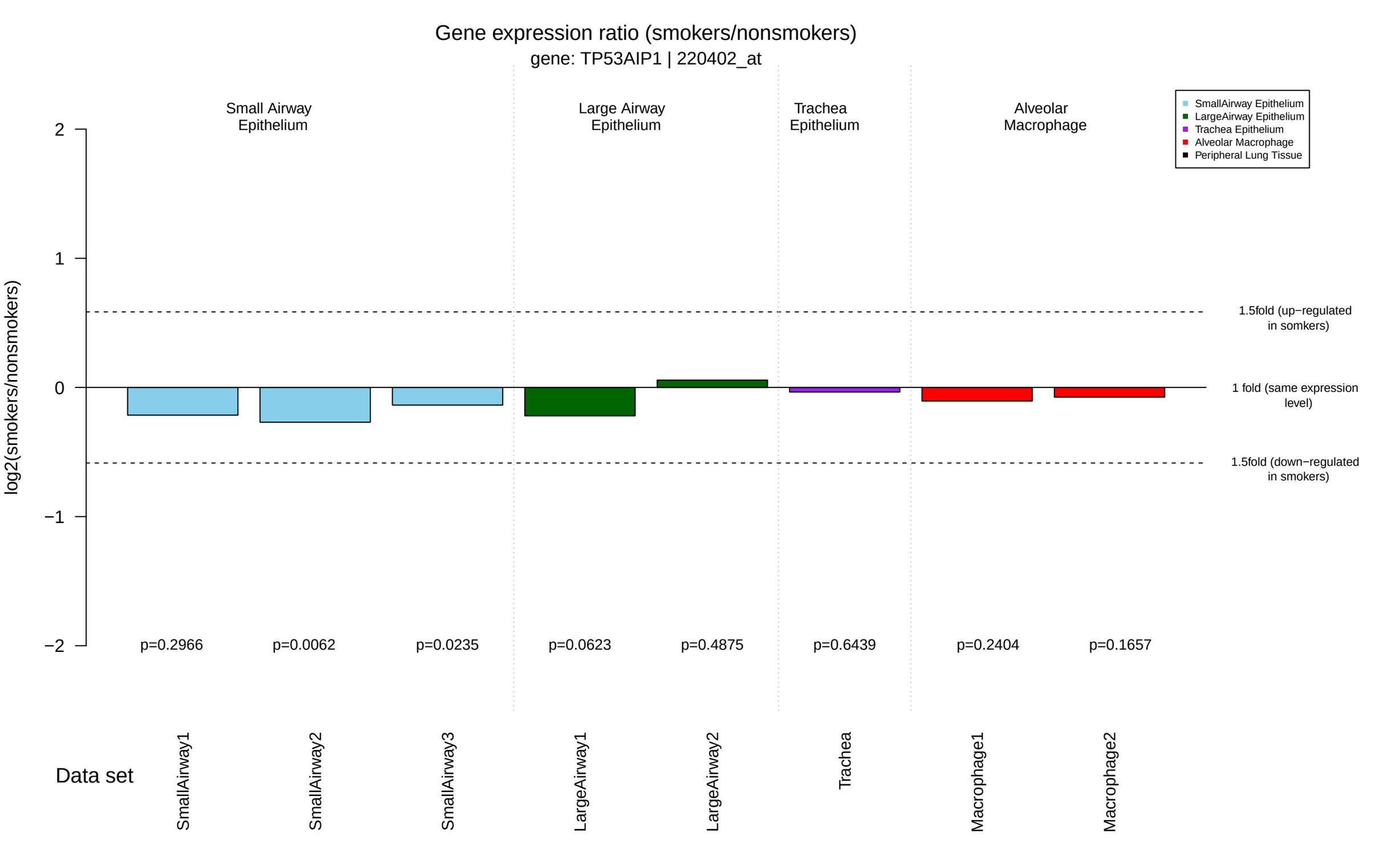
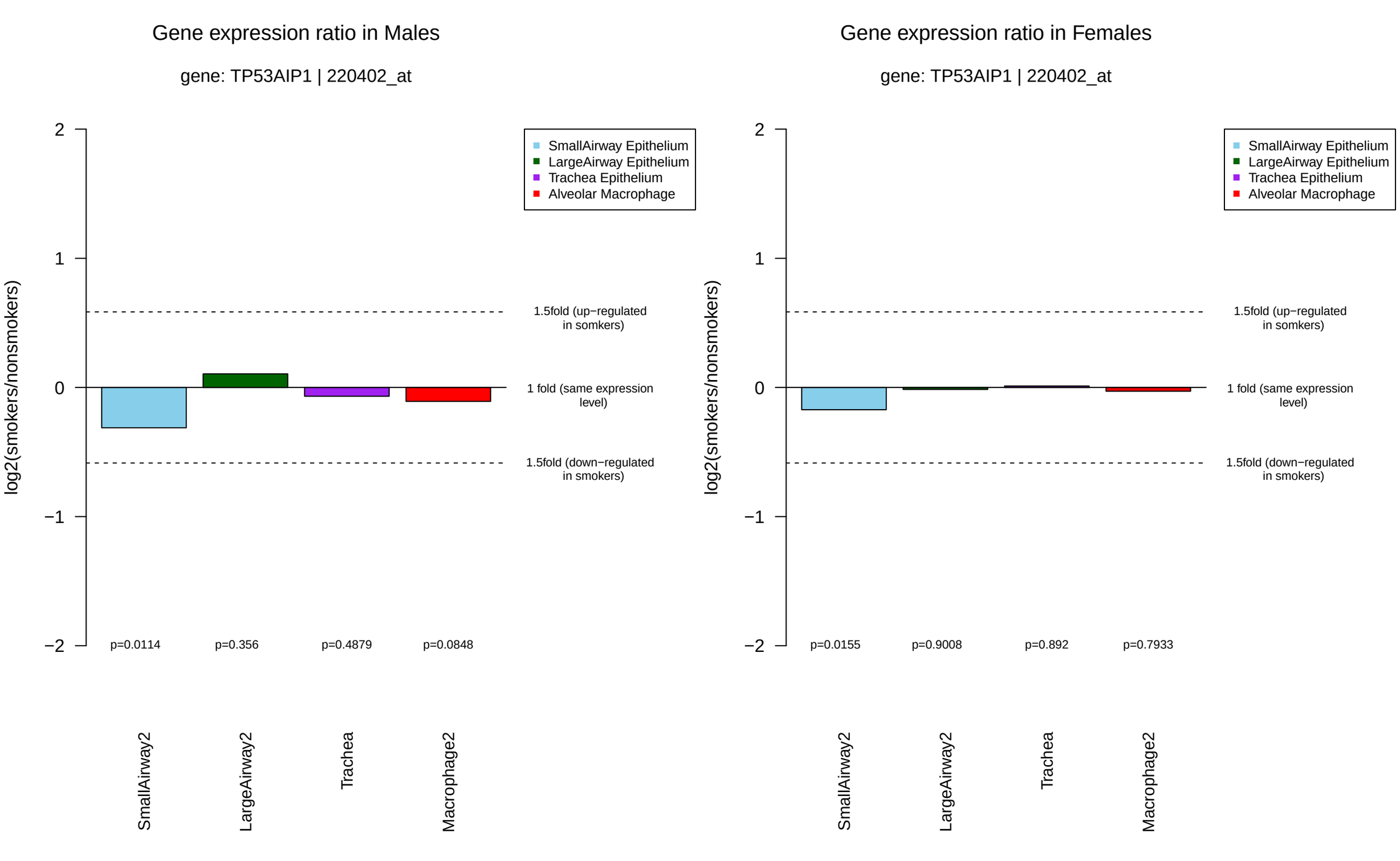
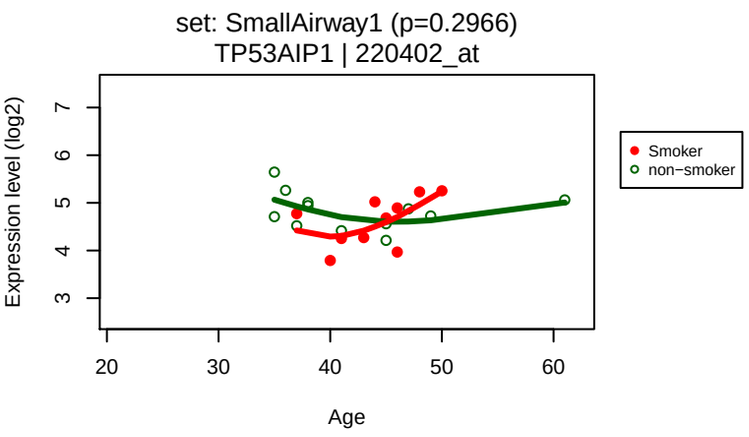
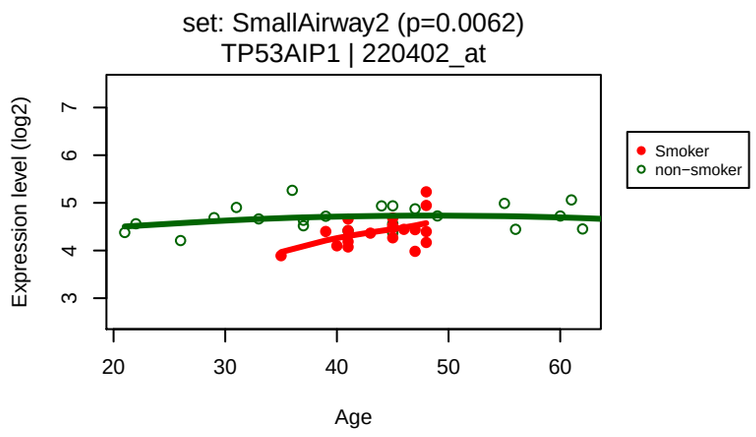
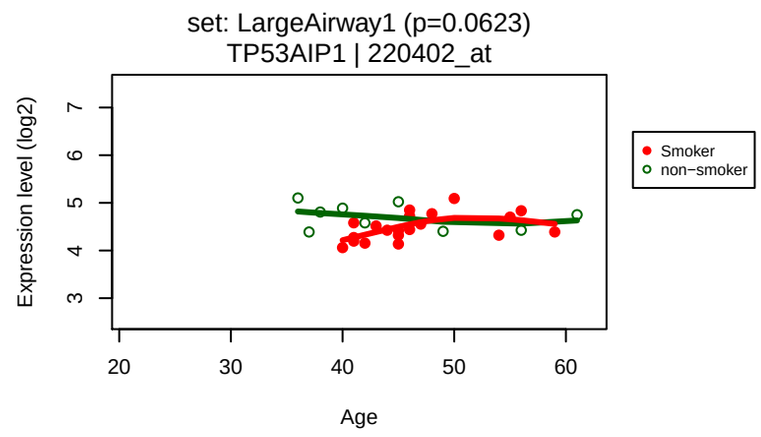

Gene expression of smokers and nonsmokers
gene: TP53AIP1 | 220402_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2966
p=0.0062
p=0.0235
p=0.0623
p=0.4875
p=0.6439
p=0.2404
p=0.1657
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

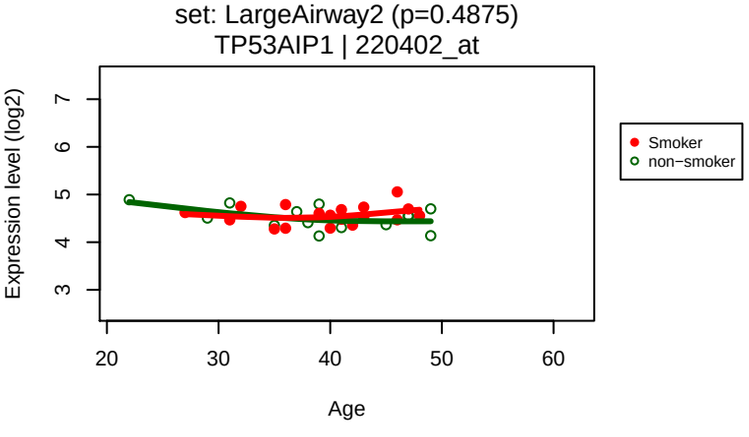

Gene expression ratio (smokers/nonsmokers)
gene: TP53AIP1 | 220402_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2966
p=0.0062
p=0.0235
p=0.0623
p=0.4875
p=0.6439
p=0.2404
p=0.1657
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

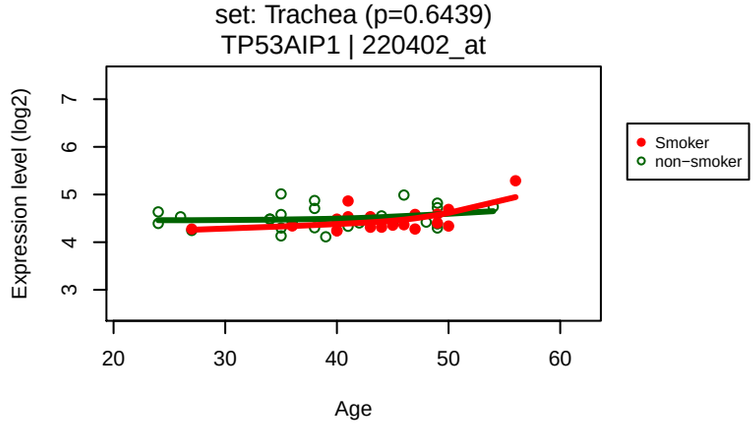

Gene expression ratio in Males
gene: TP53AIP1 | 220402_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0114
p=0.356
p=0.4879
p=0.0848
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TP53AIP1 | 220402_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0155
p=0.9008
p=0.892
p=0.7933
SmallAirway2
LargeAirway2
Trachea
Macrophage2

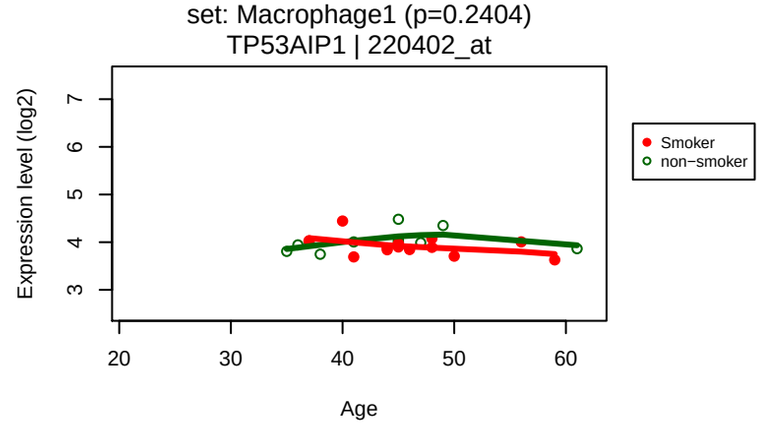

set: SmallAirway1 (p=0.2966)
TP53AIP1 | 220402_at
Expression level (log2)
Age
Smoker
non−smoker

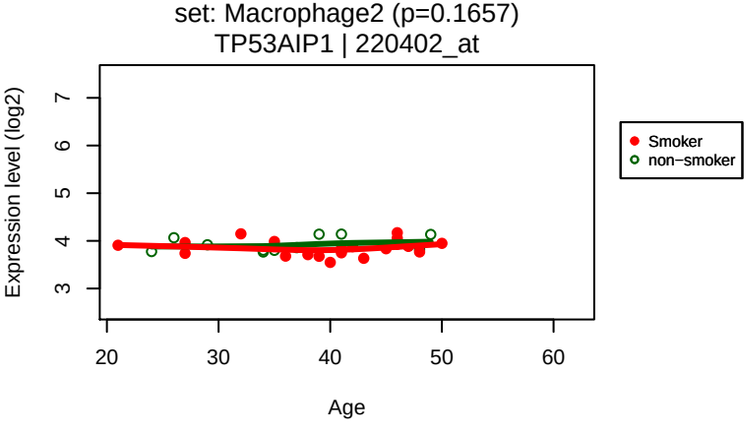

set: SmallAirway2 (p=0.0062)
TP53AIP1 | 220402_at
Expression level (log2)
Age
Smoker
non−smoker

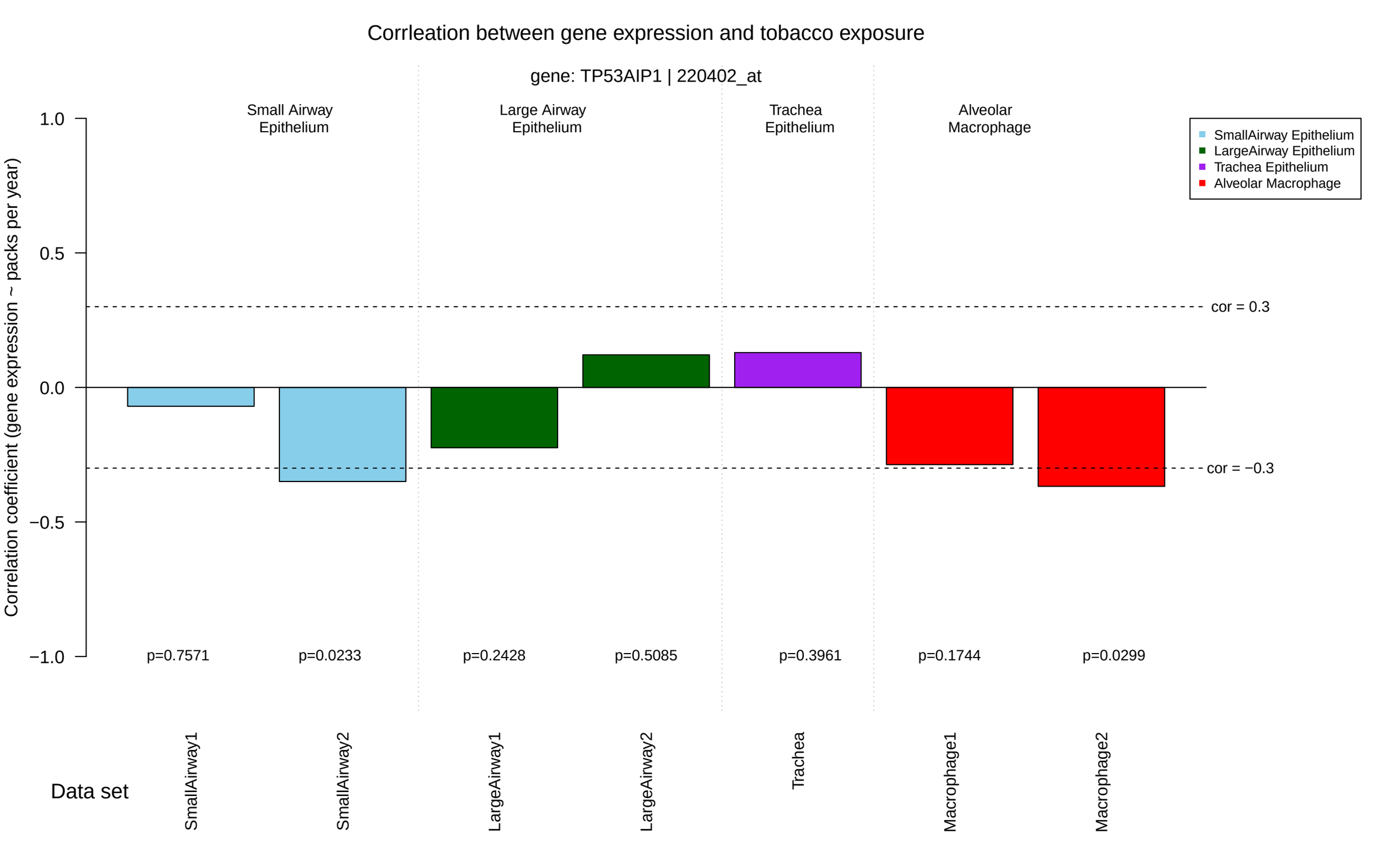

set: LargeAirway1 (p=0.0623)
TP53AIP1 | 220402_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4875)
TP53AIP1 | 220402_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6439)
TP53AIP1 | 220402_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2404)
TP53AIP1 | 220402_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1657)
TP53AIP1 | 220402_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TP53AIP1 | 220402_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7571
p=0.0233
p=0.2428
p=0.5085
p=0.3961
p=0.1744
p=0.0299
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2